Ammar Ahmed 20L-0961 29th of September 2020

BDS-1B

Day / Date

Question 1: 7 6 5 4 3 2 1 0

a) 0 1 1 1 0 1 0 1

$2^6 + 2^5 + 2^4 + 2^2 + 2^0 = \boxed{117}$

b) 1 0 1 1 1 0 1 1

$2^7 + 2^5 + 2^4 + 2^3 + 2^1 + 2^0 = \boxed{187}$

c 0 1 0 0 0 1 0 0

$2^6 + 2^2 = \boxed{68}$

d) 1 0 1 0 0 0 0 0

$2^7 + 2^5 = \boxed{160}$

e) 0 0 0 0 1 1 1 1

$2^2 + 2^2 + 2^1 + 2^0 = \boxed{15}$

Question 2: the range will be 0-255 -> $2^8 - 1$

Question 3: What is your FIRST NAME?

Question 4:

20456	50 48 52 53
196	49 57 54
1024	49 48 50 52
32	51 50
100015	49 48 48 48 49 53

Question 5 20456

01001111 11101000

2	127 - 1
2	63 - 1
2	18 - 1
2	7 - 1
2	2 - 1
	1 - 0

2	20456
2	10228 - 0
2	5114 - 0
2	2557 - 0
2	1278 - 1
2	639 - 1
2	319 - 1
	158 - 1

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196 00000000 11000100

1024

0000 0100 0000 0000

32 0000 0000 0010 0000

100015 can not be represented by 2 bytes

2	1024 - 0
	512 - 0
	256 - 0
	128 - 0
	64 - 0
	32 - 0
	16 - 0
	8 - 0
	4 - 0
	2 - 0
	1 - 1

2	196 - 0
	98 - 0
	49 - 1
	24 - 0
	12 - 0
	6 - 0
	3 - 1
	1 - 01

2	32 - 0
	16 - 0
	8 - 0
	4 - 0
	2 - 0
	1 - 0

What is the max unsigned integer value that can be represented using 2 bytes?

$(2^{16}-1)$ giving us 65535

Question 6

20456 00000000 00000000 01001111 11101000

196 00000000 00000000 00000000 11000100

1024 00000000 00000000 00000100 00000000

32 00000000 00000000 00000000 01000000

100015

00000000 00000001 10000110 10101111

2	100015 - 1
	50007 - 1
	25003 - 1
	12501 - 1
	6250 - 0
	3125 - 1
	1562 - 0

	781 - 1
	390 - 0
	195 - 1
	97 - 1
	48 - 0
	24 - 0
	12 - 0
	6 - 0
	3 - 1
	1 - 1

What is the max unsigned integer value that can be represented using 4 bytes?

$2^{16\times 2} = 32 \rightarrow 2^{32}-1$

$\nearrow$ 4294967295

Day / Date

Question 7

65536 out of 1 byte range

−64 2's compliment of 64 $\boxed{11000000}$

64 $\boxed{01000000}$

−110 110 = 0 ~~[illegible]~~ 11 2's compliment of 110

01101110 = 10010010

2	110 − 0
	55 − 1
	27 − 1
	13 − 1
	6 − 0
	3 − 1
	1 − 1
	0

Max signed 1 Byte integer = $(2^{8-1}-1) = 127$

Question 8

12 0000 1100 2's compliment method 1111 0100

−12 2's compliment of −12 → ~~1111 0100~~ 0000 1100

1111 0100

−53 53 = 00110101

2's compliment of 50 11001011

2	53 − 1
	26 − 0
	13 − 1
	6 − 0
	3 − 1
	1 − 1
	0

30 $2+2^2+2^3+2^4 =$ 0001 1110

2's compliment of 30 = 1110 0010

110 = ~~[illegible]~~ 0110 110

−110 2's compliment of 110 = 10010010

Question 9

Decimal	33	117	431
Binary	100001	1110101	0001 1010 1111
Hexadecimal	21	75	1AF

2	33 − 1
	16 − 0
	8 − 0
	4 − 0
	2 − 0
	1 − 1

00100001

2 1

7 5

0 1110101

$2^6+2^5+2^4+2^2+2^0 = 117$

1 A F

0001 1010 1111

$2^8+2^7+2^5+2^3+2^2+2^1+2^0 = 431$

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Question 2 $(6742)_{10}$ → $(\ldots)_2$

10

2	6742 – 0	2	1121 – 1
↓	3371 – 1	↓	210 – 0
	1685 – 1		105 – 1
	842 – 0		52 – 0

2	26 – 0
↓	13 – 1
	6 – 0
	3 – 1
	1 – 1

$\underbrace{0001}_{1}\ \underbrace{1010}_{A}\ \underbrace{0101}_{5}\ \underbrace{0110}_{6}$

$= \boxed{(1A56)_{16}}$

b) a) $\underbrace{1110}_{E}\ \underbrace{1110}_{E}$

b) 1010001

$\underbrace{0101}_{5}\ \underbrace{0001}_{1}$

c) 2's complement of

a) $+13$

2	13 – 1
	6 – 0
	3 – 1
	1 – 1

$(13)_{10} = \boxed{(1101)_2}$

2's complement of 13 = $\boxed{0111}$

b) -17

2	17 – 1
	8 – 0
	4 – 0
	2 – 0
	1 – 1

$\boxed{0001\ 0001}$

2's complement of 17 is

$-17 = \boxed{1110\ 1111}$

c) -55

2	55 – 1
	27 – 1
	13 – 1
	6 – 0
	3 – 1
	1 – 1

$55 = \boxed{0011\ 0111}$

$-55 = \boxed{1100\ 1001}$

d) Decimal representation of

a) $(100111011)_2$

$\underbrace{0001}\ \underbrace{0011}\ \underbrace{1011}$

$2^8 + 2^5 + 2^4 + 2^3 + 2 + 1 = 315$

b) $(B268)_{16}$

$\underbrace{1011}\ \underbrace{0010}\ \underbrace{0110}\ \underbrace{1000}$

$2^{15} + 2^{13} + 2^{12} + 2^9 + 2^6 + 2^5 + 2^3 = 45672$

e)

$01000100 = 2^6 + 2^2 = 68 = D$

$0100\ 1111 = 2^6 + 2^3 + 2^2 + 2^1 + 1 = 79 = O$

$0010\ 0000 = 2^5 = [\text{Space}]$

$0100\ 1110 = 2^6 + 2^3 + 2^2 + 2^1 = 78 = N$

$0100\ 1111 = 79 = O$

$0101\ 0100 = 2^6 + 2^4 + 2^2 = 84 = T$

$0010\ 0000 = [\text{Space}]$

$0100\ 0011 = 2^6 + 2 + 1 = 67 = C$

$0100\ 1000 = 2^6 + 2^3 = 72 = H$

$0100\ 0101 = 2^6 + 2^2 + 1 = 69 = E$

$0100\ 0001 = 2^6 + 1 = A$

$0101\ 0100 = T$

Decoded message

DO NOT CHEAT

INSPIRE